

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: FY B. Tech(All)

Branch: All Branches

Semester : I

Subject Code & Name: ENGINEERING MATHEMATICS-I

Course Code: BTBS101

Max Marks: 60

Date:21/07/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	If 5 is eigen value of the matrix A then eigen value of A^2 is.....	Remember (CO1)	1
	a. 5 b. -5 c. 25 d. 2		
2	The rank of matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$	Remember (CO1)	1
	a.3 b. 2 c. 1 d. 4		
3	Homogenous system of linear equations is	Remember (CO1)	1
	a. always consistent b. always inconsistent c. both a and b d. None of these		
4	If $u = f\left(\frac{y}{x}\right)$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ is equal to	Understand (CO2)	1
	a. 2u b. u c. 4u d. 0		
5	If $u = x^3 + y^3$ then $\frac{\partial^2 u}{\partial x \partial y}$ is equal to	Understand (CO2)	1
	a. 0 b. 3 c. 3y d. $3x^2$		
6	A point at which $f(x, y)$ has neither maxima nor minima is called as	Remember(CO3)	1
	a. Maxima b. Minima c. Saddle point d. Extreme point		
7	The curve $x^2 - 2y = c$ is symmetrical about	Remember (CO4)	1
	a. x-axis b. y-axis c. $y = x$ d. $y = -x$		
8	The point of intersection with x-axis for the curve $xy^2 = 4a^2(2a - x)$	Remember (CO4)	1
	a. (0,0) b. (2a, 0) c. (0,2a) d. (2a, 2a)		
9	If $u = x^2 - y^2$ and $v = 2xy$ then $\frac{\partial(u,v)}{\partial(x,y)}$ is equal to	Remember (CO3)	1
	a. $4(x^2 + y^2)$ b. 4x c. 4y d. None of these		

10	The number of loops for curve $r = a \sin 3\theta$ are				Remember (CO4)	1
	a. 2	b. 3	c. 6	d. 5		
11	The value of $\int_0^2 \int_0^y 5 \, dx dy$ is.				Understand (CO5)	1
	a. 2	b. 0	c. 5	d. 10		
12	The value of $\int_0^3 \int_0^2 \int_0^1 dx dy dz$.				Understand (CO5)	1
	a. 6	b. 3	c. 2	d. 1		
Q. 2	Solve the following.					12
A)	Find the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ -1 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$ by converting it into normal form.				Understand (CO1)	6
B)	Solve the system of linear equation by testing the consistency $x + y + z = 6$, $x - y + 2z = 5$, $3x + y + z = 8$, $2x - 2y + 3z = 7$				Apply(CO1)	6
Q.3	Solve the following.					12
A)	If $u = \sin^{-1}(x^2 + y^2)$ then show that $x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = 2 \tan u (2 \sec^2 u - 1)$				Understand (CO2)	6
B)	If $u = f(y - z, z - x, x - y)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$				Understand(CO2)	6
Q. 4	Solve Any Two of the following.					12
A)	Find the maxima or minima for the function $x^2 + y^2 + (30 - x - y)^2$				Understand(CO3)	6
B)	Find the maximum value of $x^2 + y^2 + z^2$ when $x + y + z = 3a$				Understand(CO3)	6
C)	Find the Jacobian $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ of $u = x^2 - 2y$, $v = x + y + z$, $w = x - 2y + 3$				Understand(CO3)	6
Q.5	Solve Any Two of the following.					12
A)	Identify and trace the curve $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$.				Apply(CO4)	6
B)	Identify and trace the curve $r = 2(1 - \cos \theta)$				Apply(CO4)	6
C)	Solve $\int_0^2 x^3 \sqrt{2-x} dx$				Apply(CO4)	6
Q. 6	Solve Any Two of the following.					12
A)	Solve $\iint xy^2 dx dy$ over the region bounded by $y = x^2$, $y = 0$, $x = 1$				Apply (CO5)	6
B)	Solve $\int_0^2 \int_0^x \int_0^{x+y} e^{x+y+z} dx dy dz$				Apply(CO5)	6
C)	Find the inverse of the matrix A by Cayley-Hamilton theorem $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$				Understand(CO1)	6
*** End ***						